

MECHATRONICS AND IOT

ASSIGNMENT 3

**Design and develop a Worker Health and
Safety Monitoring System for Construction
Industries**

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Worker Health and Safety Monitoring System for Construction Industries

1. Project Overview

Construction workers are exposed to several hazards such as excessive temperature, harmful gases, physical stress, and emergency situations. A worker health and safety monitoring system can continuously monitor important parameters and provide an immediate warning when an unsafe condition is detected.

The proposed system uses an ESP8266 NodeMCU as the main controller. A Pulse Sensor monitors the worker's heart-rate indication, a DHT11 monitors temperature and humidity, a gas sensor detects unsafe gas conditions, and a GPS module provides the worker's location. An OLED display shows the current safety status, while a buzzer provides a local warning.

Main Objectives

- Monitor worker safety parameters continuously.
- Detect unsafe environmental conditions.
- Monitor heart-rate indication.
- Provide immediate local alerts.
- Display sensor information on OLED.
- Obtain worker location using GPS.
- Develop a low-cost wearable safety prototype.

2. Problem Statement

Construction workers are frequently exposed to **unsafe environmental and physical conditions** during their daily activities. Construction sites may contain high temperatures, excessive humidity, dust, fumes, harmful gases, heavy machinery, and physically demanding tasks. Prolonged exposure to these conditions can affect worker comfort, health, concentration, and overall safety.

One major concern is **high temperature and humidity**. Workers performing physically demanding activities under hot environmental conditions may experience excessive fatigue and reduced working efficiency. Without continuous environmental monitoring, an unsafe increase in temperature or humidity may not be identified immediately.

- **High temperature exposure:** Excessive environmental temperature can increase worker discomfort and physical strain.
- **High humidity:** Excessive humidity can reduce the body's ability to regulate heat effectively.
- **Harmful gas exposure:** Gas leakage or accumulation may create an unsafe working environment.
- **Physical stress:** Heavy construction activities can place significant physical demands on workers.
- **Abnormal pulse-rate indication:** Significant changes in pulse rate may indicate that the worker requires attention.

3. Proposed Solution

The proposed system uses multiple sensors connected to an ESP8266.

Sensors used

- Pulse Sensor – monitors heart-rate indication
- DHT11 – measures temperature and humidity
- Gas Sensor – detects gas/smoke conditions
- GPS Module – provides worker location

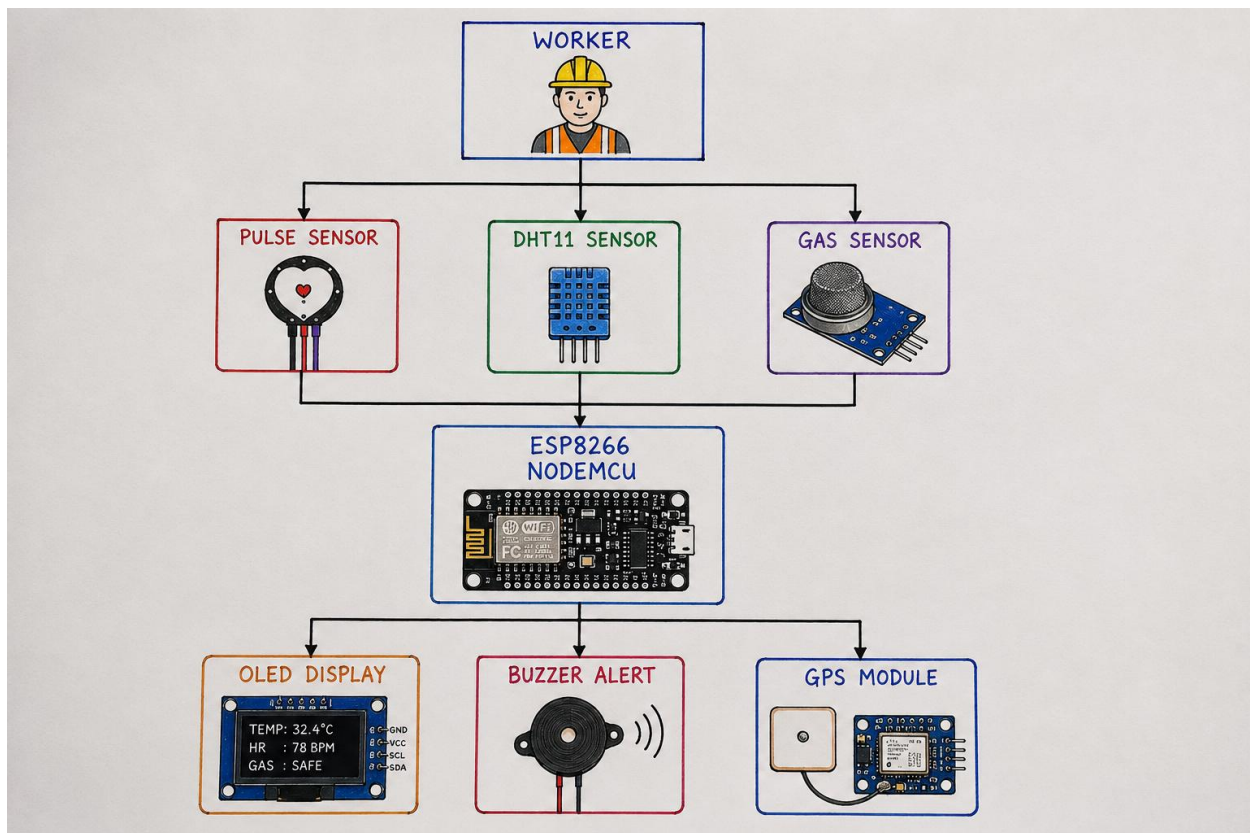
Output devices

- OLED Display – displays sensor readings and safety status
- Buzzer – generates an alarm during unsafe conditions.
- The ESP8266 continuously reads sensor values and compares them with predefined prototype thresholds.

Main Functions – Simple Explanation

1. **Pulse Sensor** – **Checks heart rate**
Measures the worker's pulse-rate indication to identify unusual readings.
2. **DHT11** – **Checks temperature and humidity**
Measures the temperature and humidity around the worker.
3. **Gas Sensor** – **Detects harmful gases/smoke**
Detects unsafe gas or smoke conditions in the working area.
4. **GPS Module** – **Finds worker location**
Provides the approximate location of the worker during an emergency.
5. **ESP8266** – **Controls the whole system**
Collects data from all sensors, processes it, and decides whether the condition is safe or unsafe.

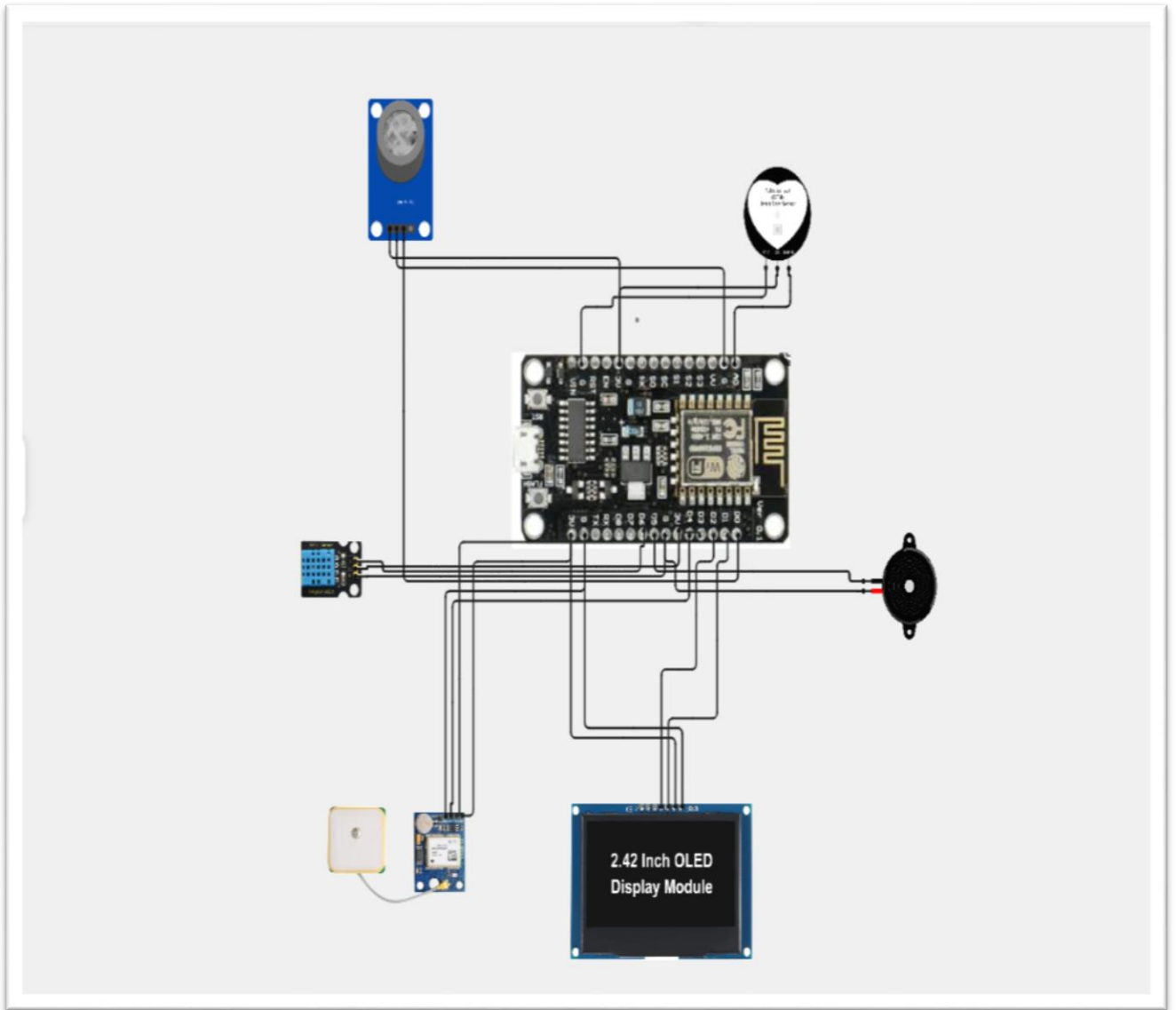
4. System Architecture



5. Circuit Connection Table

Component	Component Pin	ESP8266 NodeMCU Pin
Pulse Sensor	Signal	A0
Pulse Sensor	VCC	3.3V
Pulse Sensor	GND	GND
DHT11	DATA	D6
DHT11	VCC	3.3V
DHT11	GND	GND
Gas Sensor	DO	D0
Gas Sensor	VCC	5V/VIN*
Gas Sensor	GND	GND
OLED	SDA	D2
OLED	SCL	D1
OLED	VCC	3.3V
OLED	GND	GND
GPS	TX	D3
GPS	RX	D4
GPS	VCC	As required by module
GPS	GND	GND
Buzzer	+ (Positive)	D5
Buzzer	– (Negative)	GND

6. Circuit Design



7. Algorithm

1. Start and Initialize ESP8266:

Begin the program by powering up and initializing the main ESP8266 microcontroller.

2. Initialize Peripherals:

Set up the OLED display, DHT11 (temperature and humidity) sensor, and GPS module so they are ready to process data.

3.Read Pulse Sensor:

Acquire the current heart rate or pulse data from the connected pulse sensor.

4.Read Temperature & Humidity:

Acquire the current climate readings from the DHT11 sensor.

5.Read Gas Sensor:

Acquire the current air quality or gas level readings from the gas sensor.

6.Read GPS Data:

Collect the current geographical location coordinates from the GPS module.

7.Compare Values with Thresholds:

Evaluate all the gathered sensor readings against your predefined safety limits in the code.

8.Check Conditions (Decision Node):

Determine if all conditions are within the safe parameters:

If YES: Display **SAFE** on the OLED screen and move forward.

If NO: Turn the **Buzzer ON**, display **ALERT** on the OLED screen, and obtain the exact GPS location.

9.Repeat Loop:

Loop back to **Step 3** (Read pulse sensor) to continuously monitor the system.

8. Coding

```
#include <Wire.h>
```

```
#include <Adafruit_GFX.h>
```

```
#include <Adafruit_SSD1306.h>
```

```
#include <DHT.h>
```

```
#include <TinyGPS++.h>
```

```
#include <SoftwareSerial.h>
```

```
#define DHTPIN D6
#define DHTTYPE DHT11
#define PULSE_PIN A0
#define GAS_PIN D0
#define BUZZER_PIN D5
#define GPS_RX D3
#define GPS_TX D4
#define SCREEN_WIDTH 128
#define SCREEN_HEIGHT 64
#define OLED_RESET -1
DHT dht(DHTPIN, DHTTYPE);
Adafruit_SSD1306 display(
  SCREEN_WIDTH,
  SCREEN_HEIGHT,
  &Wire,
  OLED_RESET
);
TinyGPSPlus gps;
SoftwareSerial gpsSerial(GPS_RX, GPS_TX);
float temperature = 0;
float humidity = 0;
int pulseValue = 0;
int heartRate = 0;
bool gasAlert = false;
bool temperatureAlert = false;
bool heartAlert = false;
unsigned long lastBeat = 0;
bool beatDetected = false;
```



```
#define HIGH_TEMP 38.0
#define LOW_HR 50
#define HIGH_HR 120
#define PULSE_THRESHOLD 550

void setup()
{
  Serial.begin(115200);
  gpsSerial.begin(9600);
  pinMode(GAS_PIN, INPUT);
  pinMode(BUZZER_PIN, OUTPUT);
  digitalWrite(BUZZER_PIN, LOW);
  dht.begin();
  if (!display.begin(
    SSD1306_SWITCHCAPVCC,
    0x3C))
  {
    Serial.println("OLED ERROR");
    while (1);
  }
  display.clearDisplay();
  display.setTextColor(SSD1306_WHITE);
  display.setTextSize(1);
  display.setCursor(15, 20);
  display.println("WORKER SAFETY");
  display.setCursor(20, 35);
  display.println("MONITORING");
  display.display();
  delay(2000);
```

```

}
void loop()
{
  readTemperature();
  readPulse();
  readGas();
  readGPS();
  checkSafety();
  showDisplay();
  delay(100);
}
// -----
// DHT11
// -----
void readTemperature()
{
  float t = dht.readTemperature();
  float h = dht.readHumidity();
  if (!isnan(t))
    temperature = t;
  if (!isnan(h))
    humidity = h;
  temperatureAlert =
    (temperature >= HIGH_TEMP);
}
// -----
// PULSE SENSOR
// -----

```

```
void readPulse()
{
pulseValue = analogRead(PULSE_PIN);
if (pulseValue > PULSE_THRESHOLD &&
!beatDetected)
{
beatDetected = true;
unsigned long currentTime = millis();
if (lastBeat > 0)
{
unsigned long interval =
currentTime - lastBeat;
if (interval > 300 &&
interval < 1500)
{
heartRate = 60000 / interval;
}
}
lastBeat = currentTime;
}
if (pulseValue <
PULSE_THRESHOLD - 30)
{
beatDetected = false;
}
if ((heartRate > 0 &&
heartRate <= LOW_HR) ||
heartRate >= HIGH_HR)
```

```
{
heartAlert = true;
}
else
{
heartAlert = false;
}
}
// -----
// GAS SENSOR
// -----
void readGas()
{
int gasState = digitalRead(GAS_PIN);

if (gasState == LOW)
{
gasAlert = true;
}
else
{
gasAlert = false;
}
}

// -----
// GPS
// -----
```

```

void readGPS()
{
while (gpsSerial.available())
{
gps.encode(gpsSerial.read());
}
}

// -----
// SAFETY CHECK
// -----

void checkSafety()
{
if (temperatureAlert ||
heartAlert ||
gasAlert)
{
digitalWrite(BUZZER_PIN, HIGH);
Serial.println("WARNING!");
if (temperatureAlert)
Serial.println("High Temperature");
if (heartAlert)
Serial.println("Abnormal Heart Rate");
if (gasAlert)
Serial.println("Gas Detected");
if (gps.location.isValid())
{
Serial.print("Latitude: ");
Serial.println(gps.location.lat(), 6);

```

```
Serial.print("Longitude: ");
Serial.println(gps.location.lng(), 6);
}
}
else
{
digitalWrite(BUZZER_PIN, LOW);
}
}
// -----
// OLED
// -----
void showDisplay()
{
display.clearDisplay();
display.setTextSize(1);
display.setTextColor(SSD1306_WHITE);
display.setCursor(15, 0);
display.println("WORKER SAFETY");
display.setCursor(0, 14);
display.print("Temp: ");
display.print(temperature, 1);
display.println(" C");
display.setCursor(0, 25);
display.print("Humidity: ");
display.print(humidity, 0);
display.println("%");
display.setCursor(0, 36);
```

```
display.print("Heart: ");
if (heartRate > 0)
{
display.print(heartRate);
display.println(" BPM");
}
else
{
display.println("-- BPM");
}
display.setCursor(0, 47);
display.print("Gas: ");
if (gasAlert)
display.println("DANGER");
else
display.println("SAFE");
display.setCursor(75, 47);
if (temperatureAlert ||
heartAlert ||
gasAlert)
{
display.println("ALERT");
}
else
{
display.println("SAFE");
}
display.display();
```

9. System Working



Step 1 – Sensor initialization

The ESP8266 initializes:

- Pulse sensor
- DHT11
- Gas sensor
- GPS
- OLED
- Buzzer

Step 2 – Data collection

The sensors continuously provide data to the ESP8266.

Pulse → Heart-rate indication

DHT11 → Temperature + Humidity

Gas → Gas condition

GPS → Location

Step 3 – Data processing

The ESP8266 processes the sensor readings and compares them with the programmed prototype thresholds.

Step 4 – Safety decision

If the readings are normal:

STATUS = SAFE

If an unsafe condition occurs:

STATUS = ALERT

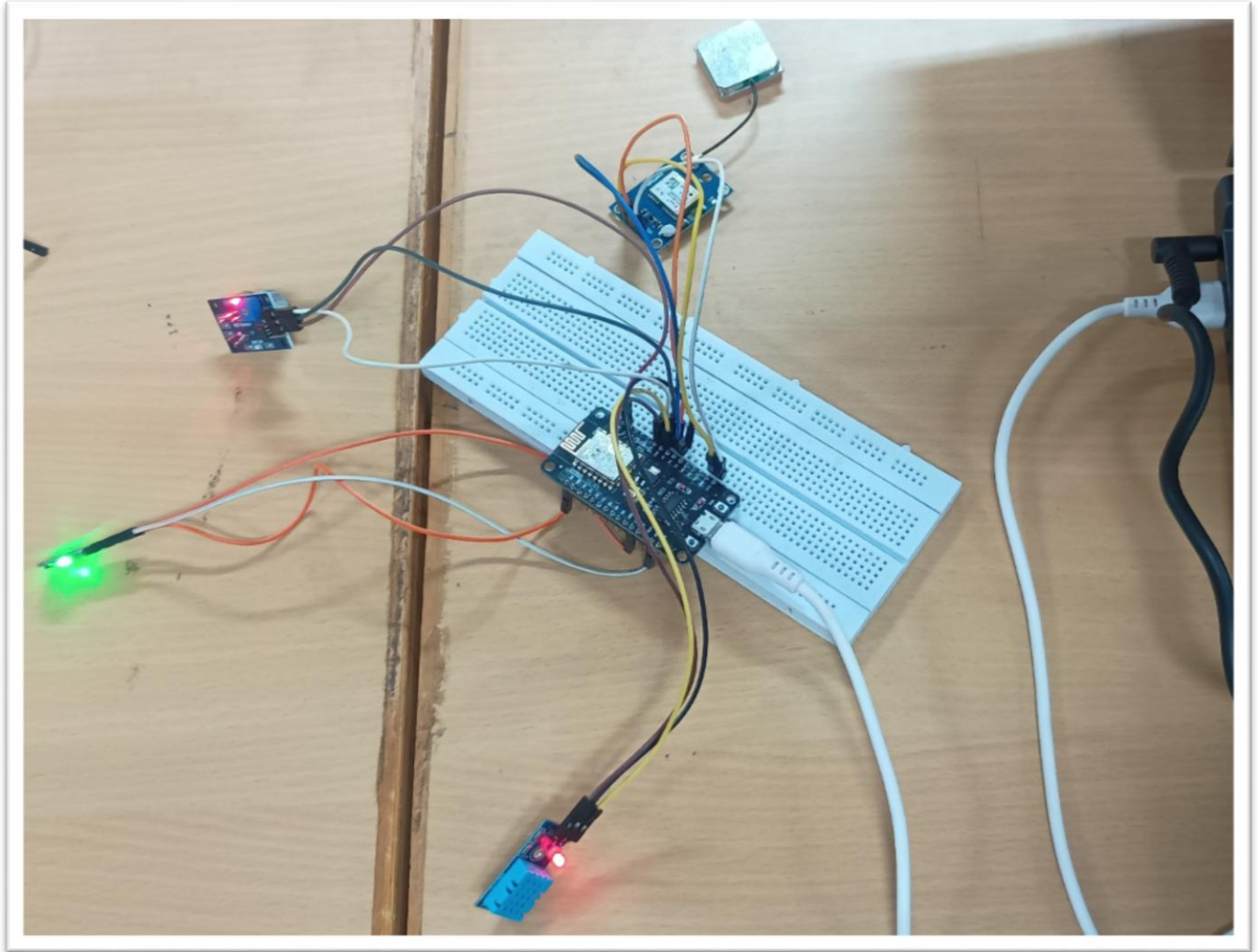
Step 5 – Alert

The buzzer is activated and the OLED displays the warning.

10. Bill of Materials

S.No	Component	Qty.	Approx. Price
1	ESP8266 NodeMCU	1	₹300–450
2	Pulse Sensor	1	₹150–300
3	DHT11	1	₹50–100
4	MQ Gas Sensor	1	₹100–180
5	GPS NEO-6M	1	₹400–600
6	2.42" OLED	1	₹300–600
7	Buzzer	1	₹20–50
8	Breadboard	1	₹80–150
9	Jumper wires	1	₹50–100
10	Battery/Power supply	1	₹150–300
11	Helmet/Prototype enclosure	1	₹100–300

11. Prototype Implementation



Prototype demonstration

You can demonstrate three conditions

Condition 1 – Normal

Temperature → Normal

Heart rate → Normal indication

Gas → Safe

OLED → SAFE
Buzzer → OFF

Condition 2 – High temperature

Temperature → Above prototype threshold

OLED → ALERT
Buzzer → ON

Condition 3 – Gas detected

Gas Sensor → Danger
OLED → GAS DANGER
Buzzer → ON
GPS → Location available

12. Results & Performance Analysis

Performance

- The **Pulse Sensor** provides real-time heart-rate indication.
- The **DHT11** successfully measures temperature and humidity.
- The **Gas Sensor** detects unsafe gas/smoke conditions.
- The **GPS module** provides the worker's location when a valid satellite signal is available.
- The **OLED display** shows sensor values and safety status.
- The **buzzer** provides an immediate warning during unsafe conditions.
- The ESP8266 processes the sensor data and makes the safety decision.
- The prototype responds quickly to programmed unsafe conditions.
- The system successfully demonstrates **SAFE** and **ALERT** modes. The system continuously monitors worker safety parameters.

Advantages

- Low-cost and easy to build.
- Real-time worker safety monitoring.
- Provides immediate local warning.

- Compact and portable.
- Can be mounted on a safety helmet or vest.
- Multiple parameters can be monitored simultaneously.
- GPS helps identify the worker's location.
- OLED provides easy-to-read information.
- Reduces dependence on manual monitoring.
- Can be expanded with additional sensors and IoT features.

Disadvantages

- DHT11 has relatively low accuracy compared with industrial temperature sensors.
- Pulse Sensor readings can be affected by finger movement and poor contact.
- Gas sensor readings can vary with environmental conditions.
- GPS may take time to obtain a location.
- GPS performance can be poor indoors or in areas with obstructions.
- The system requires a continuous power supply.
- Basic buzzer alerts only provide local warning.
- ESP8266 has limited input/output resources compared with more powerful controllers.

Limitations

- The prototype is **not a medical-grade health monitoring device**.
- Heart-rate readings should be treated as an indication, not a medical diagnosis.
- The gas sensor provides an indication of unsafe gas conditions rather than precise industrial gas concentration measurements.
- The programmed safety thresholds are prototype values and should not be used as official occupational-safety limits.
- The current system does not detect falls unless an accelerometer/gyroscope such as **MPU6050** is added.

